Gold Commodity Price Prediction (4-Year Analysis)

Project Report

Jan 2026

1 Executive Summary

This project builds a full, reproducible pipeline to forecast next-day gold futures movement using the latest 4 years of data and a strict time-ordered split. The holdout winner by RMSE is the **Naive (No-Change)** baseline (**96.50**), while **Ridge** is selected as the best non-naive model for recursive forward simulation.

Key numerical outcomes:

- Raw rows: 1,033; model-ready rows: 978.
- Feature count after engineering: 42.
- Holdout period: 2025-09-12 to 2026-04-14 (147 trading days).
- Latest observed close: 4,839.20.
- 30-business-day Ridge forecast endpoint: 5,096.09.
- Projected horizon change: +256.89 points (+5.31%).

2 Data Source, Scope, and Profile

Instrument: Gold Futures (`GC=F`) from Yahoo Finance
Date range: 2022-03-07 to 2026-04-15
Raw observations: 1,033 daily rows

Data is downloaded with `yfinance`, cleaned, sorted chronologically, and deduplicated. After feature generation and NA removal, 978 observations remain.

2.1 Basic Data Characteristics

Variable	Mean	Std. Dev.	Min	Max
Open	2587.29	921.08	1620.40	5415.70
High	2603.38	932.22	1623.30	5586.20
Low	2571.54	907.77	1615.10	5301.60
Close	2587.86	920.05	1623.30	5318.40
Volume	4100.13	20237.16	0.00	202373.00

Table 1: Descriptive statistics over the full 4-year sample.

3 Mathematical Formulation

3.1 Target Definition

Let C_t be the close price at time t. The one-step return target is:

$$r_{t+1} = \frac{C_{t+1} - C_t}{C_t}$$

Predicted return is converted to predicted next-day close:

$$\hat{C}_{t+1} = C_t(1 + \hat{r}_{t+1})$$

3.2 Feature Engineering Equations

Main technical features are built from price history:

$$\mathrm{SMA}_{w,t} = \frac{1}{w} \sum_{i=0}^{w-1} C_{t-i}$$

$$\mathrm{EMA}_{w,t} = \alpha C_t + (1 - \alpha)\mathrm{EMA}_{w,t-1}, \quad \alpha = \frac{2}{w+1}$$

$$\sigma_{w,t} = \sqrt{\frac{1}{w-1} \sum_{i=0}^{w-1} (r_{t-i} - \bar{r}_{w,t})^2}$$

$$\mathrm{RSI}_{14,t} = 100 - \frac{100}{1 + \mathrm{RS}_t}, \quad \mathrm{RS}_t = \frac{\mathrm{AvgGain}_{14,t}}{\mathrm{AvgLoss}_{14,t} + \varepsilon}$$

$$\mathrm{MACD}_t = \mathrm{EMA}_{12,t} - \mathrm{EMA}_{26,t}, \quad \mathrm{Signal}_t = \mathrm{EMA}_9(\mathrm{MACD}_t)$$

Calendar seasonality is encoded cyclically:

$$\mathrm{dow_sin}_t = \sin\left(\frac{2\pi \cdot \mathrm{dow}_t}{7}\right), \quad \mathrm{dow_cos}_t = \cos\left(\frac{2\pi \cdot \mathrm{dow}_t}{7}\right)$$

$$\mathrm{month_sin}_t = \sin\left(\frac{2\pi \cdot \mathrm{month}_t}{12}\right), \quad \mathrm{month_cos}_t = \cos\left(\frac{2\pi \cdot \mathrm{month}_t}{12}\right)$$

4 Train/Validation/Test Protocol

Chronological splitting is used to prevent leakage:

Split	Rows	Features	Date Range
Train	684	42	2022-05-23 to 2025-02-11
Validation	147	42	2025-02-12 to 2025-09-11
Test (Holdout)	147	42	2025-09-12 to 2026-04-14

Table 2: Strict chronological split used across all models.

5 Models and Objective Functions

5.1 Naive Baseline

$$\hat{r}_{t+1} = 0 \quad \Rightarrow \quad \hat{C}_{t+1} = C_t$$

5.2 Ridge Regression

Ridge is trained with standardized features and $\lambda = \alpha = 5.0$:

$$\hat{\beta} = \arg\min_{\beta} \left(\|y - X\beta\|_2^2 + \lambda \|\beta\|_2^2 \right)$$

5.3 Random Forest Regressor

Random Forest prediction is the average over $M = 700$ trees:

$$\hat{r}_{t+1} = \frac{1}{M} \sum_{m=1}^{M} T_m(x_t)$$

Configured with `max_depth=10`, `min_samples_leaf=2`, `random_state=42`.

6 Evaluation Metrics

All metrics are computed on reconstructed close prices:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} |C_i - \hat{C}_i|, \quad \text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (C_i - \hat{C}_i)^2}$$

$$\text{MAPE}(\%) = \frac{100}{n} \sum_{i=1}^{n} \left| \frac{C_i - \hat{C}_i}{C_i} \right|, \quad R^2 = 1 - \frac{\sum_{i=1}^{n} (C_i - \hat{C}_i)^2}{\sum_{i=1}^{n} (C_i - \bar{C})^2}$$

$$\text{Directional Accuracy}(\%) = \frac{100}{n} \sum_{i=1}^{n} \mathbf{1}\{\text{sign}(r_i) = \text{sign}(\hat{r}_i)\}$$

7 Model Results

7.1 Validation Performance

Model	MAE	RMSE	MAPE (%)	R^2	Directional Acc. (%)
Naive (No-Change)	30.85	41.01	0.95	0.9537	57.14
Ridge	31.22	42.21	0.96	0.9509	49.66
RandomForest	32.68	43.96	1.01	0.9468	40.14

Table 3: Validation-set results (lower is better for MAE/RMSE/MAPE).

On validation, Naive leads RMSE; Ridge is +2.92% RMSE versus Naive, and Random Forest is +7.20%.

7.2 Holdout Test Performance

Model	MAE	RMSE	MAPE (%)	R^2	Directional Acc. (%)
Naive (No-Change)	64.10	96.50	1.41	0.9542	60.54
Ridge	65.48	97.65	1.44	0.9531	51.02
RandomForest	67.98	99.94	1.50	0.9509	48.98

Table 4: Holdout test results (lower is better for MAE/RMSE/MAPE).

Interpretation:

- Best holdout RMSE: **Naive (96.50)**.
- Ridge is +1.19% RMSE versus Naive; Random Forest is +3.57%.
- Directional accuracy drops by 9.52 percentage points (Ridge) and 11.56 points (Random Forest) vs Naive.
- This indicates strong short-horizon persistence in gold prices, making a no-change benchmark hard to beat.

7.3 Bias and Error Dispersion on Holdout

Model	Mean Error (Bias)	MAE	Error Std. Dev.
Naive (No-Change)	-8.09	64.10	96.49
Ridge	+6.61	65.48	97.76
RandomForest	-20.02	67.98	98.25

Table 5: Error profile using $\hat{C}_i - C_i$ on holdout test.

8 30-Business-Day Forecast (Ridge)

Recursive forecasting updates history with each predicted point:

$$\hat{C}_{t+h} = \hat{C}_{t+h-1}(1 + \hat{r}_{t+h}), \quad h = 1, \ldots, 30$$

Forecast Statistic	Value
Horizon length	30 business days
Forecast start date	2026-04-16
Forecast end date	2026-05-27
First predicted close	4869.53
Final predicted close	5096.09
Minimum predicted close	4869.53
Maximum predicted close	5096.09
Mean predicted close	4957.29
Mean daily predicted return	0.1727%
Min/Max daily predicted return	-0.2532% / +0.6267%
Absolute projected change	+256.89
Projected change (%)	+5.31%

Table 6: Numerical summary of the recursive Ridge forecast path.

9 Visual Outputs

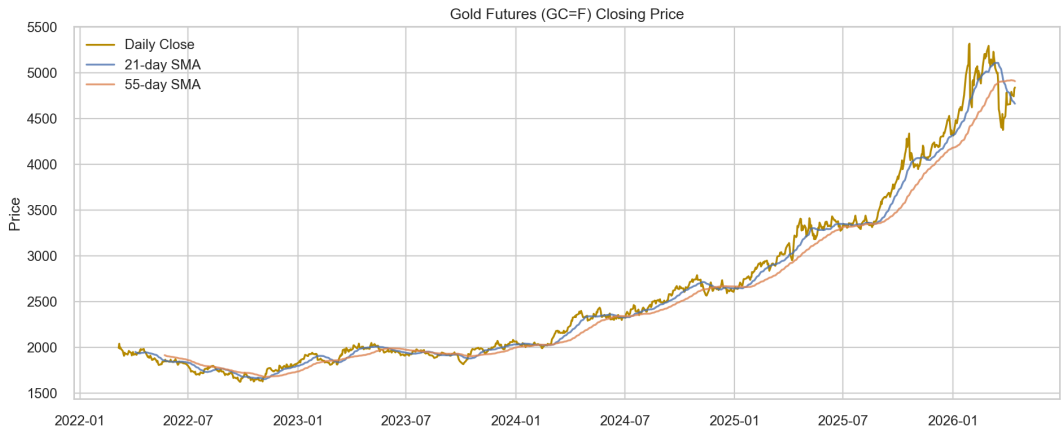

Figure 1: Gold futures close history with moving averages.

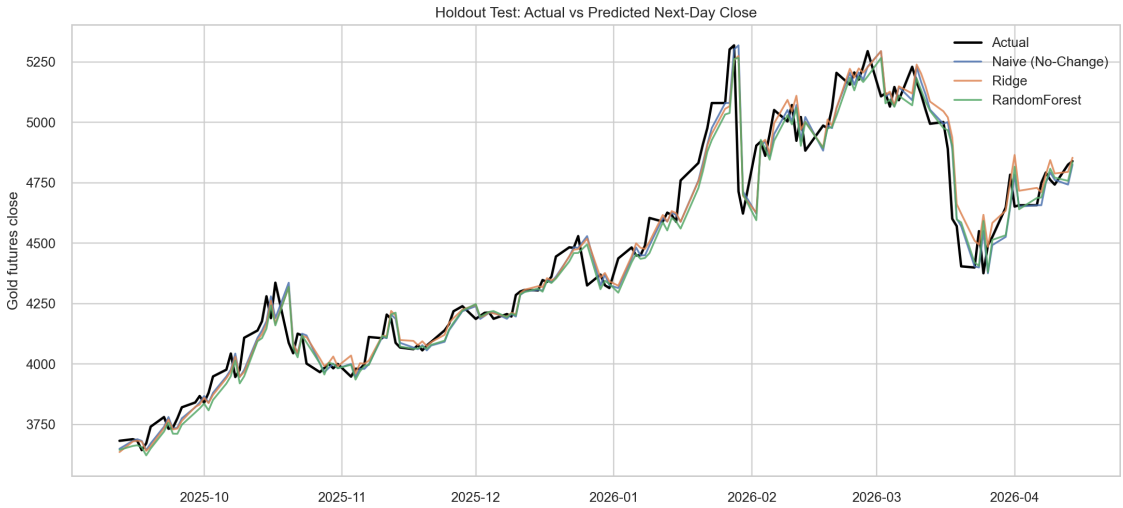

Figure 2: Holdout test: actual vs predicted next-day close (all models).

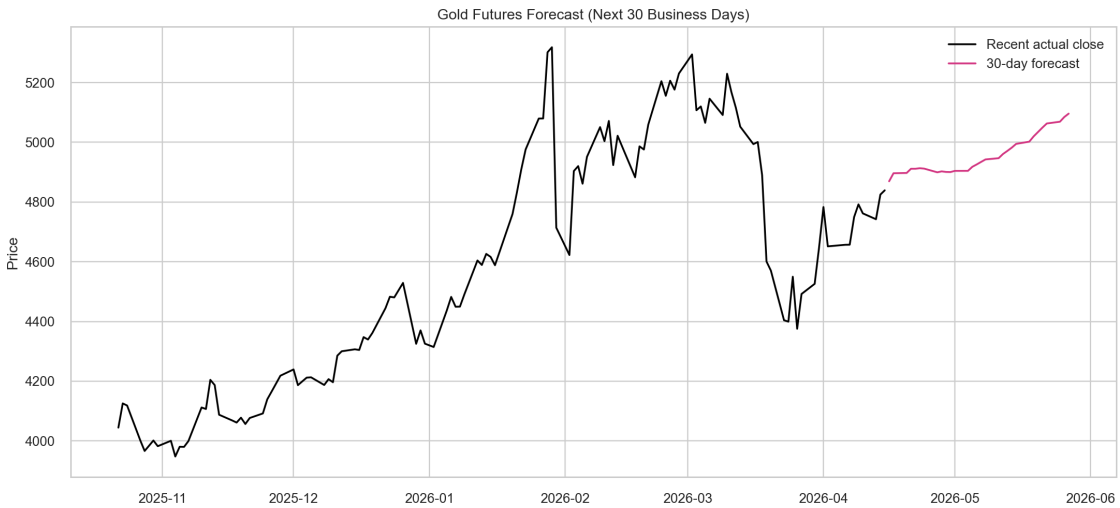

Figure 3: 30-business-day recursive forecast path generated by Ridge.

10 Conclusion

The pipeline is complete and reproducible, with clear benchmarked evidence across validation and holdout splits. The key empirical result is that a persistence baseline remains strongest at one-day horizon in this sample, while Ridge provides a structured non-naive alternative for forward scenario generation.

11 Reproducibility and Deliverables

Main project outputs:

- `gold_price_prediction_4y_complete.ipynb`
- `project_report.pdf`
- `artifacts/metrics_summary.json`
- `artifacts/validation_results.csv`
- `artifacts/test_results.csv`
- `artifacts/test_predictions.csv`
- `artifacts/future_forecast.csv`
- `artifacts/gold_close_history.png`
- `artifacts/test_actual_vs_pred.png`
- `artifacts/forecast_next_30d.png`